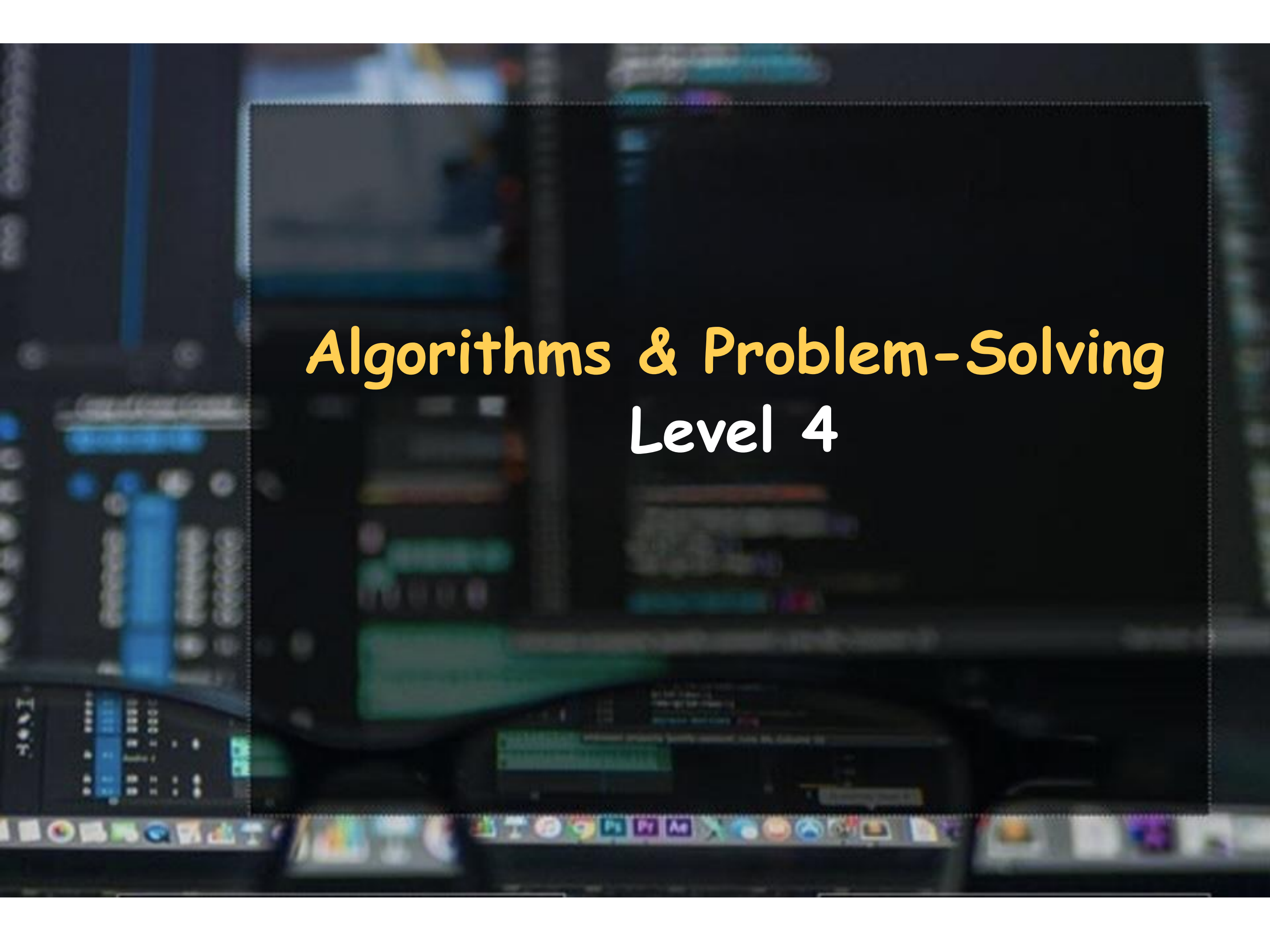
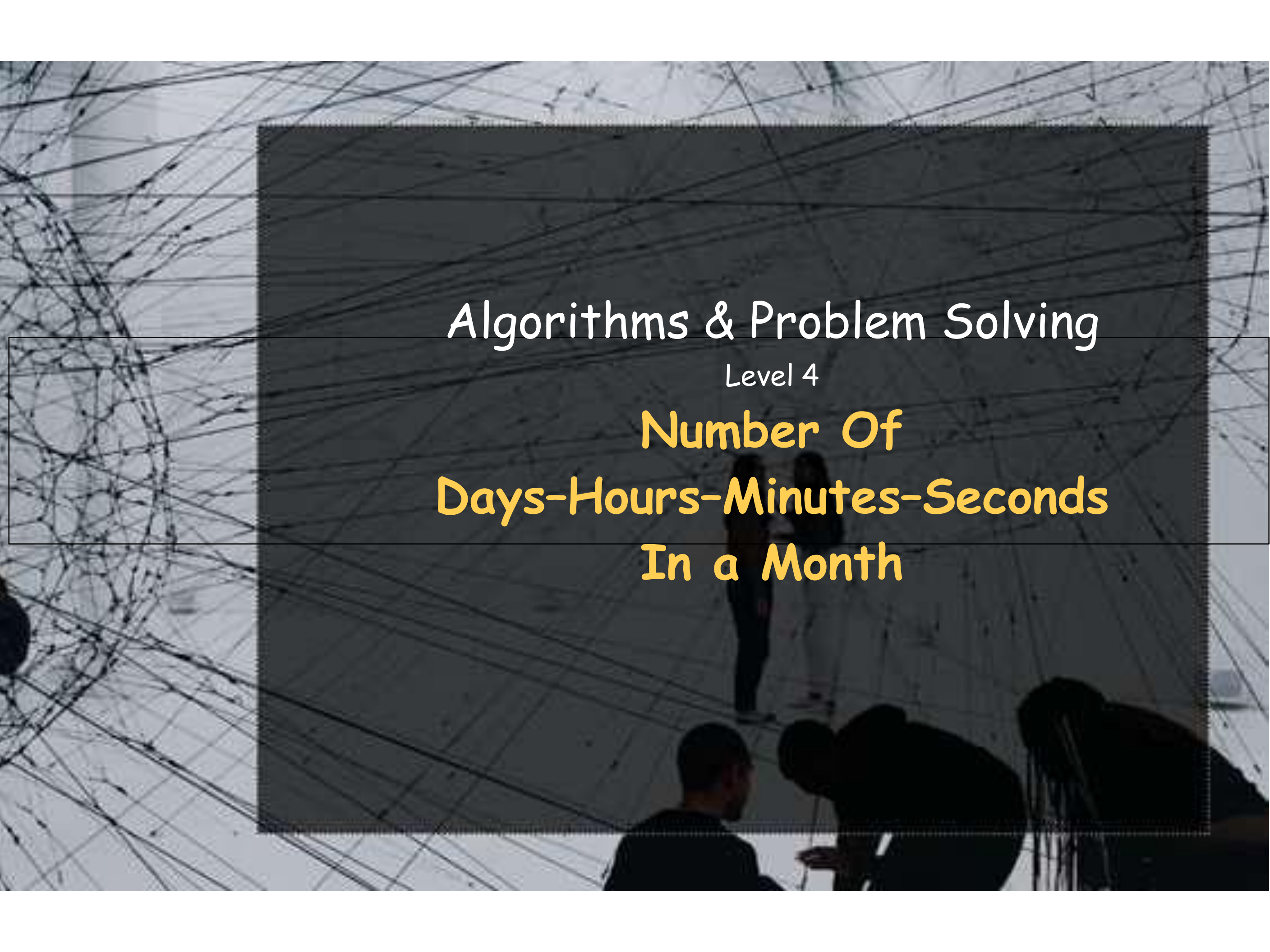




Algorithms & Problem-Solving Level 4

The background of the slide features a complex, abstract pattern of thin, dark lines crisscrossing over a lighter, textured surface. In the lower portion of the image, there are dark silhouettes of several people, some standing and some appearing to be in a group, which adds a human element to the technical-looking background.

Algorithms & Problem Solving

Level 4

Number Of Days-Hours-Minutes-Seconds In a Month



Definition

- The goal is a program to calculate the number of days in a specific month of a given year, using concise logic (one line to determine the number of days).



Explanation

- The program relies on simple and direct logic. Instead of using multiple conditions, the number of days for each month is stored in an array, and a three-part condition is used to determine if the month is February (2) and to determine if the year is a leap year.

Procedures

① The ReadYear() function

- Requires the user to enter the year and returns it.

② The ReadMonth() function

- Requires the user to enter the month number and returns it.

③ The IsLeapYear(int year) function

- Checks if the year is a leap year.

- A leap year is one that:

- (divisible by 4 but not by 100) or (divisible by 400).

④ The NumberOfDaysInAMonth(int month, int year) function

- Uses an array:

- ```
int NumberOfDays[12] = {31, 28, 31, 30, 31, 30, 31, 30, 31, 30, 31};
```

- Then returns the value based on the triple condition:

- ```
return(month == 2) ? (IsLeapYear(year) ? 29 : 28) : NumberOfDays[month - 1];
```

⑤ The main() function

- Reads the year and month and displays the number of days directly

Example:

Please enter a year to check? 2000

Please enter a Month to check? 2

Number of Days in Month [2] is 29

Note:

 If the month number is less than 1 or greater than 12, the value 0 is returned.

Summary

- ✓ The program relies on:
 - Arrays
 - Term Operators
 - Functions

Thank you 😊